

OAKLAND INTL, CA, USA

WMO: 724930

Lat: 37.7444N

Lon: 122.2236W

Elev: 3

StdP: 101.29

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23230

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.7	4.0	-6.6	2.2	11.6	-3.5	2.8	10.9	11.8	11.7	10.5	11.9	1.4	50	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	9.2	28.8	18.0	26.4	17.3	24.1	16.7	19.2	25.9	18.4	24.3	17.7	23.0	4.8	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	12.2	20.5	16.2	11.5	20.0	15.4	10.9	19.4	54.4	25.9	51.7	24.4	49.6	23.0	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	8.9	8.1	DB	0.3	34.3	1.4	2.1	-0.7	35.8	-1.5	37.0	-2.2	38.2	-3.2	39.7	
			WB	-1.1	21.0	1.5	1.2	-2.2	21.8	-3.1	22.5	-3.9	23.2	-5.0	24.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	10.3	11.5	13.0	14.1	15.5	17.0	17.7	18.1	18.4	16.6	13.0	10.4
	DBStd	3.67	2.26	2.28	2.29	2.40	2.29	2.29	1.88	2.00	2.51	2.46	2.53	2.47
	HDD10.0	66	23	10	3	1	0	0	0	0	0	0	5	24
	HDD18.3	1446	248	191	166	129	95	53	36	28	29	65	159	246
	CDD10.0	1764	33	52	96	124	171	209	237	252	251	206	96	37
	CDD18.3	102	0	0	1	3	7	13	14	21	29	12	1	0
	CDH23.3	671	0	1	6	29	60	107	69	94	200	99	4	0
	CDH26.7	192	0	0	1	5	17	35	17	27	67	23	0	0
Wind	WSAvg	3.8	2.9	3.5	3.9	4.4	4.7	4.5	4.5	4.2	3.7	3.2	2.8	3.1
Precipitation	PrecAvg	454	93	70	71	31	6	3	1	1	6	32	69	67
	PrecMax	901	230	182	204	138	61	30	19	22	38	208	200	154
	PrecMin	184	7	3	0	0	0	0	0	0	0	0	3	0
	PrecStd	174	67	54	50	34	11	7	4	4	9	41	52	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	20.9	23.9	27.1	28.9	31.1	29.0	30.3	32.2	29.4	23.4	18.6
		MCWB	10.7	14.0	14.6	16.4	17.2	18.5	19.3	19.0	18.6	16.7	15.0	11.7
	2%	DB	16.5	18.3	21.0	22.9	24.4	26.3	25.1	25.8	28.3	26.1	20.7	16.9
		MCWB	12.2	12.7	14.1	14.9	16.1	17.4	18.1	18.1	18.0	15.8	14.0	12.2
	5%	DB	15.1	16.8	18.6	20.2	21.4	23.1	22.7	23.1	25.3	23.4	18.6	15.6
		MCWB	11.7	12.1	13.2	14.1	15.3	16.5	17.0	17.4	17.3	15.3	13.6	12.2
	10%	DB	13.9	15.4	17.2	18.0	19.2	21.2	21.3	21.9	22.7	21.3	17.3	14.4
		MCWB	11.3	12.1	12.7	13.3	14.4	15.8	16.4	16.9	16.9	15.1	13.3	11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	15.5	16.4	17.6	18.9	20.1	20.0	20.0	20.2	18.7	16.9	15.4
		MCDB	15.8	17.6	22.0	24.2	26.1	27.0	27.4	27.5	27.7	25.0	21.2	16.7
	2%	WB	13.6	14.3	15.1	15.9	17.0	18.2	18.3	18.6	19.0	17.5	15.7	14.4
		MCDB	14.9	16.8	18.7	20.7	22.5	24.6	24.5	24.2	25.6	22.7	18.6	15.6
	5%	WB	12.7	13.4	14.2	14.9	15.8	17.0	17.3	17.8	18.1	16.7	14.9	13.4
		MCDB	14.0	15.8	17.2	19.0	20.3	22.6	22.3	22.8	23.8	21.5	17.4	14.7
	10%	WB	11.9	12.6	13.4	14.0	14.9	16.0	16.6	17.0	17.3	15.8	14.1	12.3
		MCDB	13.3	14.7	16.3	17.5	18.8	20.7	20.9	21.3	22.2	20.1	16.5	13.7
Mean Daily Temperature Range	5% DB	MDBR	7.3	7.5	8.2	8.3	7.7	8.3	7.6	7.8	9.2	9.6	8.4	7.4
		MCDBR	8.9	9.7	11.4	12.2	12.1	12.7	10.5	10.5	15.1	13.9	10.7	8.3
		MCWBR	5.3	5.6	6.2	6.3	5.7	5.8	4.8	4.9	6.4	6.7	6.1	5.2
	5% WB	MCDBR	6.0	7.9	9.1	10.6	10.6	11.5	9.9	10.4	12.3	11.1	8.2	5.2
		MCWBR	4.3	5.0	5.5	5.9	5.4	5.6	4.7	4.9	5.7	5.8	5.3	4.0
Clear-Sky Solar Irradiance	taub	0.324	0.330	0.342	0.360	0.364	0.360	0.369	0.349	0.348	0.340	0.324	0.325	
	taud	2.549	2.532	2.484	2.401	2.417	2.435	2.413	2.465	2.454	2.521	2.548	2.551	
	Ebn at Noon	864	905	922	920	916	916	905	919	903	879	856	836	
	Edh at Noon	76	88	101	116	116	114	116	107	102	87	76	72	
All-Sky Solar Radiation	RadAvg	2.38	3.32	4.70	6.15	7.10	7.77	7.37	6.58	5.56	4.09	2.81	2.09	
	RadStd	0.34	0.40	0.42	0.49	0.46	0.33	0.24	0.28	0.21	0.26	0.18	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	-1.19	N/A	N/A	N/A	N/A	-110	+135	N/A

Nomenclature: See separate page